EXPERIMENT – 5 Alarm Based Motor ON/OFF System

PROJECT NAME

Alarm Based Motor ON/OFF System Using Arduino UNO, DS1307 RTC Module and SSD1306 OLED Display

---

OBJECTIVE / PROBLEM STATEMENT

To design and implement an automatic motor ON/OFF control system using Arduino UNO. The system detects water supply using a switch sensor and controls a pump indicator LED based on water availability.

The DS1307 RTC module is used for real-time tracking and timing operations. The SSD1306 OLED display continuously displays real time, water supply status, and pump status.

The system also checks whether water supply continues for more than one hour.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
DS1307 RTC Module	1
SSD1306 OLED Display	1
Push Button / Water Sensor Switch	1
LED (Pump Indicator)	1
Jumper Wires	Required
Breadboard	1

PIN CONNECTIONS

WATER SENSOR CONNECTION

Device	Arduino Pin
Water Supply Switch	D2

PUMP LED CONNECTION

Device	Arduino Pin
Pump LED	D3

DS1307 RTC CONNECTIONS

RTC Pin	Arduino Pin
VCC	5V
GND	GND
SDA	A4
SCL	A5

SSD1306 OLED DISPLAY CONNECTIONS

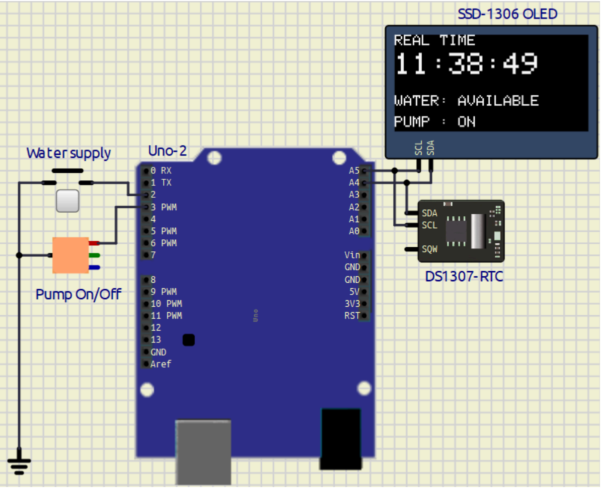
OLED Pin	Arduino Pin
VCC	5V
GND	GND
SDA	A4
SCL	A5

SOFTWARE USED

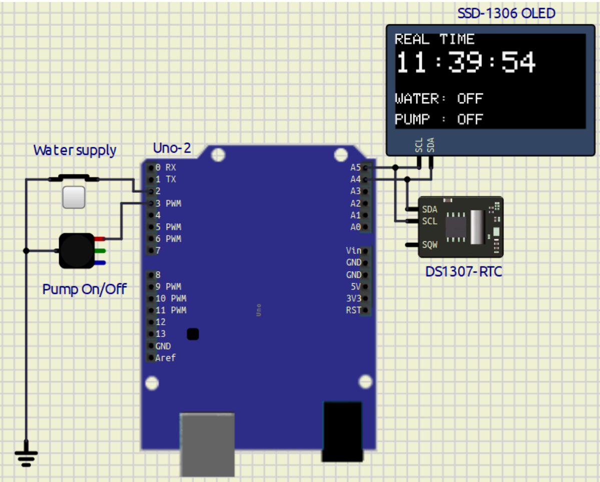
- Arduino IDE
- SimulIDE / Proteus

CIRCUIT DIAGRAM

1. When Water supply available



2. When water supply is off



REQUIRED LIBRARIES

```
#include <Wire.h>
#include <RTClib.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
```

MAIN ARDUINO CODE

Code With Comment ----> Click Here

```
/*
   =====================================================
      Alarm Based Motor ON/OFF System using RTC
   =====================================================
*/
```

```cpp
#include <Wire.h>
#include <RTClib.h>

#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>


// OLED SETTINGS

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH,
                         SCREEN_HEIGHT,
                         &Wire,
                         -1);


// RTC OBJECT

RTC_DS1307 rtc;


// PIN DEFINITIONS

#define WATER_SENSOR 2
#define PUMP_LED     3


// VARIABLES

bool pumpState = false;

bool timerStarted = false;


// Time when water supply started

DateTime waterStartTime;



// SETUP

void setup()
{
  pinMode(WATER_SENSOR, INPUT_PULLUP);

  pinMode(PUMP_LED, OUTPUT);

  digitalWrite(PUMP_LED, LOW);
```

```
  Wire.begin();

  rtc.begin();


  // Uncomment first time only
  // rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));


  // OLED START

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
  {
    while (1);
  }

  display.clearDisplay();
  display.display();

  delay(1000);
}



// MAIN LOOP

void loop()
{
  DateTime now = rtc.now();

  bool waterAvailable = digitalRead(WATER_SENSOR);


  // WATER AVAILABLE

  if (waterAvailable == LOW)
  {
    // Turn Pump ON

    digitalWrite(PUMP_LED, HIGH);

    pumpState = true;


    // Start Timer Only Once

    if (!timerStarted)
    {
      waterStartTime = now;

      timerStarted = true;
    }

```

```
      // Calculate Time Difference

      TimeSpan elapsed = now - waterStartTime;


      // Check 1 Hour Condition

      if (elapsed.totalseconds() >= 3600)
      {
        // Additional logic can be added here
      }
    }



    // WATER NOT AVAILABLE

    else
    {
      timerStarted = false;


      // Wait 2 Seconds

      delay(2000);

      digitalWrite(PUMP_LED, LOW);

      pumpState = false;
    }



    // UPDATE OLED DISPLAY

    displayData(now, waterAvailable);


    delay(200);
  }



  // OLED DISPLAY FUNCTION

  void displayData(DateTime now, bool waterAvailable)
  {
    display.clearDisplay();

    display.setTextSize(1);

    display.setTextColor(WHITE);

```

```
// REAL TIME

display.setCursor(0, 0);

display.println("REAL TIME");

display.setTextSize(2);

display.setCursor(0, 12);

print2digit(now.hour());

display.print(":");

print2digit(now.minute());

display.print(":");

print2digit(now.second());



// WATER STATUS

display.setTextSize(1);

display.setCursor(0, 40);

display.print("WATER: ");

if (waterAvailable == LOW)
{
  display.println("AVAILABLE");
}
else
{
  display.println("OFF");
}



// PUMP STATUS

display.setCursor(0, 54);

display.print("PUMP : ");

if (pumpState)
{
  display.println("ON");
}
else
{
```

```
    display.println("OFF");
  }


  display.display();
}



// PRINT 2 DIGIT NUMBER

void print2digit(int number)
{
  if (number < 10)
  {
    display.print("0");
  }

  display.print(number);
}
```

WORKING PRINCIPLE

1. Water sensor continuously checks water supply.
2. If water is available:
   - Pump LED turns ON.
   - RTC timer starts.
3. If water supply continues for more than 1 hour:
   - Timer condition becomes true.
4. If water supply stops before 1 hour:
   - System waits 2 seconds.
   - Pump LED turns OFF.
5. OLED continuously displays:
   - Real Time
   - Water Supply Status
   - Pump Status

OUTPUT

OLED Display

```
REAL TIME
12:45:30

WATER: AVAILABLE
```

```
PUMP : ON
```

Water Supply OFF

```
REAL TIME
12:46:10

WATER: OFF

PUMP : OFF
```

FEATURES

- Automatic pump control
- Real-time clock display
- OLED status monitoring
- Water detection system
- 1-hour timing condition
- I2C communication
- Simple embedded automation system

ADVANTAGES

- Easy to implement
- Low-cost project
- Real-time monitoring
- Automatic control system
- Useful for water management systems

APPLICATIONS

- Water tank automation
- Motor protection systems
- Smart irrigation systems
- Industrial timing systems
- Embedded automation projects

CONCLUSION

The Alarm Based Motor ON/OFF System using Arduino UNO, DS1307 RTC module, and SSD1306 OLED display was successfully implemented and tested. The system correctly detects water supply conditions, controls pump operation, and displays real-time status information on the OLED display.

During this experiment, I learned:

- RTC interfacing
- OLED interfacing
- I2C communication
- Digital input/output control
- Embedded automation logic

REFERENCES

1. Arduino UNO Official Documentation
   https://docs.arduino.cc/hardware/uno-rev3/

2. DS1307 RTC Datasheet
   https://datasheets.maximintegrated.com/en/ds/DS1307.pdf

3. RTClib Library
   https://github.com/adafruit/RTClib

4. Adafruit SSD1306 Library
   https://github.com/adafruit/Adafruit_SSD1306

5. Adafruit GFX Library
   https://github.com/adafruit/Adafruit-GFX-Library

6. Arduino Wire Library Documentation
   https://www.arduino.cc/en/reference/wire